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EDA Project

Supply Chain Analysis

Final Report

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SUPERVISOR'S CERTIFICATE

This is to certify that the work reported in the B. Tech Dissertation/dissertation proposal entitled "**Supply Chain Analysis"**, submitted by **KOTHAPU SANTOSH KUMAR REDDY** at **Lovely Professional University, Phagwara, India** is a Bonafide record of his / her original work carried out under my supervision. This work has not been submitted elsewhere for any other degree.

Signature of Supervisor

Hritwiz Yash

Date:

Acknowledgement

I hereby declare that the research work reported in the dissertation/dissertation proposal entitled "Supply Chain Analysis" in partial fulfilment of the requirement for the award of Degree for Bachelor of Technology in Computer Science and Engineering at Lovely Professional University, Phagwara, Punjab is an authentic work carried out under supervision of my research supervisor **Mr. Hritwiz Yash**. I have not submitted this work elsewhere for any degree or diploma.

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1. PROBLEM STATEMENT

In the modern logistics and supply chain environment, data-driven decision-making plays a crucial role in improving operational efficiency and maximizing profitability. With the increasing complexity of supply networks, organizations must rely on analytical insights to identify inefficiencies, reduce costs, and enhance service quality. This project focuses on performing an in-depth Exploratory Data Analysis (EDA) on a supply chain dataset to uncover hidden patterns, operational bottlenecks, and key drivers of profitability.

The dataset contains transactional and operational details such as order dates, delivery timelines, warehouse information, product identifiers, pricing, discounts, and cost structures. The primary objective of this analysis is to understand how these factors influence revenue, profit margins, and delivery performance. Python libraries such as **Pandas**, **Matplotlib**, and **Seaborn** are used extensively for data manipulation, visualization, and statistical analysis.

The project begins with comprehensive **data cleaning**, including handling missing values, correcting data types, removing duplicates, and validating date fields. This ensures the reliability and consistency of the dataset. Following this, **feature engineering** is performed to derive meaningful business metrics such as total revenue, total cost, profit, profit margin, order processing time, shipping time, and total delivery time. These engineered features enable a deeper understanding of operational efficiency and financial performance.

Visual analysis is conducted using various charts and plots to identify trends and anomalies across products, customers, warehouses, and time periods. One of the key findings of the analysis is the identification of **loss-making products and transactions**, highlighting the impact of excessive discounts, high procurement costs, and long holding periods. Additionally, warehouse-level performance is evaluated to assess processing consistency.

To validate operational differences between warehouses, **statistical hypothesis testing** is performed using Independent T-tests. This helps determine whether observed delays in order processing times are statistically significant or due to random variation. The results reveal notable performance differences among warehouses, indicating potential process inefficiencies and resource imbalances.

Based on the insights obtained, the project provides **actionable recommendations** such as optimizing discount strategies, improving inventory turnover, streamlining warehouse operations, and prioritizing high-margin products and customers. Overall, this EDA project demonstrates how analytical techniques and data-driven insights can support strategic decision-making and enhance supply chain profitability, reliability, and customer satisfaction.

1.Introduction

1.1 Overview

Supply chain management refers to the coordinated management of the flow of goods, information, and services from raw material suppliers to end customers. It includes key activities such as procurement, production, warehousing, transportation, and delivery. An efficient supply chain is essential for reducing costs, improving delivery performance, and increasing customer satisfaction.

In today's competitive business environment, organizations generate large volumes of transactional data through sales, inventory, and logistics operations. Analysing this data provides valuable insights into operational performance and financial outcomes. This project utilizes historical sales and order data to analysis the efficiency of the supply chain and to understand how different operational factors influence profitability and delivery timelines. By applying Exploratory Data Analysis (EDA) techniques, the project aims to uncover patterns, trends, and inefficiencies that are not easily visible through traditional reporting methods.

1.2 Problem Statement

Many businesses face challenges in managing their supply chain due to hidden inefficiencies and lack of data-driven insights. Common issues include inconsistent shipping and delivery times, high operational and holding costs, excessive discounting, and products that consistently generate losses. These problems often remain unnoticed because organizations rely on aggregated metrics rather than detailed transactional analysis.

Without granular analysis, it becomes difficult to identify specific warehouses responsible for processing delays or to determine which products, customers, or discount strategies are negatively impacting overall profitability. As a result, decision-makers may implement broad strategies that fail to address the root causes of inefficiency. This project addresses these challenges by performing a detailed exploratory and statistical analysis to isolate problem areas within the supply chain.

1.3 Objectives

The main objectives of this project are as follows:

- To clean, preprocess, and standardize raw sales and logistics data to ensure accuracy and consistency.
- To perform feature engineering to derive key financial and operational metrics such as revenue, profit, and delivery time.
- To identify loss-making products and transactions that negatively impact profit margins.
- To statistically compare warehouse performance using hypothesis testing techniques and identify operational bottleneck.

1.4 Scope

The scope is limited to the structured dataset provided (expanded_sales_data_for_eda.csv). The analysis focuses on descriptive and diagnostic analytics using Python. It does not include predictive modelling (Machine Learning) at this stage, but lays the groundwork for future forecasting models.

2.Literature Review

2.1 Introduction

A literature review provides an overview of existing research and studies related to supply chain analytics and data-driven decision-making. It helps in understanding how analytical techniques have been applied in logistics and supply chain management to improve operational efficiency and profitability. This chapter reviews relevant studies on supply chain performance analysis, exploratory data analysis (EDA), and the use of statistical methods in identifying operational bottlenecks.

2.2 Supply Chain Performance Analysis

Several researchers have emphasized the importance of measuring supply chain performance using key metrics such as delivery time, inventory turnover, order fulfilment rate, and profit margins. According to Chopra and Meindl, effective supply chain management focuses on balancing cost efficiency with responsiveness to customer demand. Poor coordination between procurement, warehousing, and transportation often leads to increased delays and higher operational costs.

Previous studies have shown that warehouse operations play a critical role in overall supply chain efficiency. Delays at the warehouse level can significantly affect delivery timelines and customer satisfaction. Researchers suggest that continuous monitoring of warehouse performance metrics can help organizations detect inefficiencies early and improve process reliability.

2.3 Role of Exploratory Data Analysis in Supply Chains

Exploratory Data Analysis (EDA) is widely used as an initial step in data analytics projects to understand data distributions, detect outliers, and identify hidden patterns. Tukey introduced EDA as a method to summarize and visualize data before applying advanced modelling techniques. In supply chain contexts, EDA helps organizations analysis sales trends, demand variability, discount behaviour, and inventory movement.

Many studies highlight the effectiveness of EDA tools such as data visualization, correlation analysis, and segmentation in identifying loss-making products and inefficient processes. Visual analytics allow decision-makers to quickly interpret large datasets and make informed strategic decisions. EDA also supports feature engineering by enabling the creation of

meaningful business metrics from raw transactional data.

2.4 Statistical Methods in Warehouse and Logistics Analysis

Statistical hypothesis testing has been widely applied in supply chain research to compare performance across different operational units. Techniques such as T-tests and ANOVA are commonly used to determine whether differences in processing time, delivery time, or costs across warehouses are statistically significant.

Research indicates that using statistical tests helps avoid decisions based on assumptions or random variations. For example, Independent T-tests are effective in comparing average processing times between two warehouses to identify genuine performance gaps. Such methods provide quantitative evidence to support operational improvements and resource reallocation.

2.5 Research Gap

While existing literature highlights the importance of analytics in supply chain management, many studies focus on predictive modelling or optimization techniques rather than detailed exploratory analysis. There is limited emphasis on combining EDA with statistical hypothesis testing to identify loss-making products and warehouse-level bottlenecks using real-world transactional data.

This project addresses this gap by integrating exploratory data analysis with statistical testing to provide actionable insights into supply chain inefficiencies. The study focuses on practical business metrics and interpretable results that can directly support managerial decision-making.

3.METHODOLOGY

3.1 Dataset Description

The analysis in this project was performed using the dataset **USA_sales_data** source. The raw dataset consists of **100,100 rows and 16 columns**, representing transactional sales and logistics information across multiple regions and warehouses in the United States.

Each record in the dataset corresponds to a single sales transaction and captures details related to order placement, procurement, shipment, delivery, pricing, and discounting. The dataset provides a complete view of the order lifecycle, making it suitable for analyzing operational efficiency and financial performance within the supply chain.

Key Attributes

- **Categorical Attributes:**
    - Sales Channel
    - WarehouseCode
    - CurrencyCode
- **Numerical Attributes:**
    - Order Quantity
    - Discount Applied
    - Unit Cost
    - Unit Price
- **Temporal (Date) Attributes:**
    - ProcuredDate
    - OrderDate
    - ShipDate
    - DeliveryDate

These attributes allow the derivation of important business metrics such as revenue, profit, processing time, and delivery duration.

dataset used in this project is **US Regional Sales Data**, obtained from **Kaggle**. The dataset contains real-world transactional sales and logistics records related to supply chain operations across different regions and warehouses in the United States. The data source link is provided at the end of the report.

3.2 Data Preprocessing

Data preprocessing is a crucial step in Exploratory Data Analysis (EDA) as the quality of insights depends heavily on the quality of data. The raw dataset contained missing values, inconsistent formats, and unprocessed date fields. All preprocessing steps were performed using **Python in a Jupyter Notebook**, primarily leveraging the Pandas library.

The main preprocessing tasks included handling missing values, standardizing data types, and preparing the dataset for feature engineering and statistical analysis.

3.2.1 Handling Missing Values

The dataset contained a significant number of missing values, with approximately **40% missing entries in several columns**. Different imputation strategies were applied based on the nature and importance of each attribute.

- **Order_Number:**
  Rows with missing *OrderNumber* were removed from the dataset. Since the order number uniquely identifies a transaction, missing values in this column made the records unusable for analysis.

- **Categorical_Imputation:**
  Missing values in categorical columns such as *Sales Channel* and *Warehouse Code* were replaced with the label **"Unknown"**. This approach preserves the row count while allowing the model to treat missing categories as a separate group during analysis.

- **Numerical_Imputation:**
  Missing values in *Unit Cost* and *Unit Price* were filled using the **median** of their respective columns. The median was chosen instead of the mean to minimize the influence of extreme values and outliers commonly present in pricing data.

- **Date_Imputation:**

  A logic-based imputation strategy was applied to missing date values. For example, if *Order Date* was missing but *Ship Date* was available, the missing *Order Date* was estimated by subtracting the median shipping time from the *Ship Date*. This ensured chronological consistency between procurement, order, shipping, and delivery events.

3.2.2 Data Cleaning

- **Formatting:** Column names were standardized (e.g., Sales Channel to Sales_Channel).
- **Currency Conversion:** Characters like $ and , were removed from cost/price columns to convert them into floats.
- **Duplicate Removal:** 7,269 duplicate rows were identified and removed to ensure data integrity.

3.3 Feature Engineering

To extract meaningful insights, new features were created:

1. **Financial Metrics:**
   - Total_Price = Order Quantity * Unit Price
   - Total_Cost = Order Quantity * Unit Cost
   - Revenue = Total Price * (1 - Discount Applied)
   - Profit = Revenue - Total Cost
2. **Time-Based Metrics (Lead Times):**
   - Order_Processing_Time = Ship Date - Order Date
   - Shipping_Time = Delivery Date - Ship Date
   - Total_Delivery_Time = Delivery Date - Order Date

3.4 Statistical Analysis Approach

To validate operational differences between warehouses, **statistical hypothesis testing** was performed. Independent T-tests were used to compare average order processing times between selected warehouses. This approach helped determine whether observed delays were statistically significant or due to random variation.

A significance level of **0.05** was used to evaluate the test results. This statistical validation strengthens the reliability of the insights and supports data-driven decision-making.

4.DATA ANALYSIS AND INTERPRETATION

This chapter presents the results of the exploratory data analysis performed on the supply chain dataset. The analysis is divided into univariate, bivariate, and operational efficiency analyses to understand data distributions, profitability drivers, and warehouse-level performance.

4.1 Univariate Analysis: Understanding Distributions

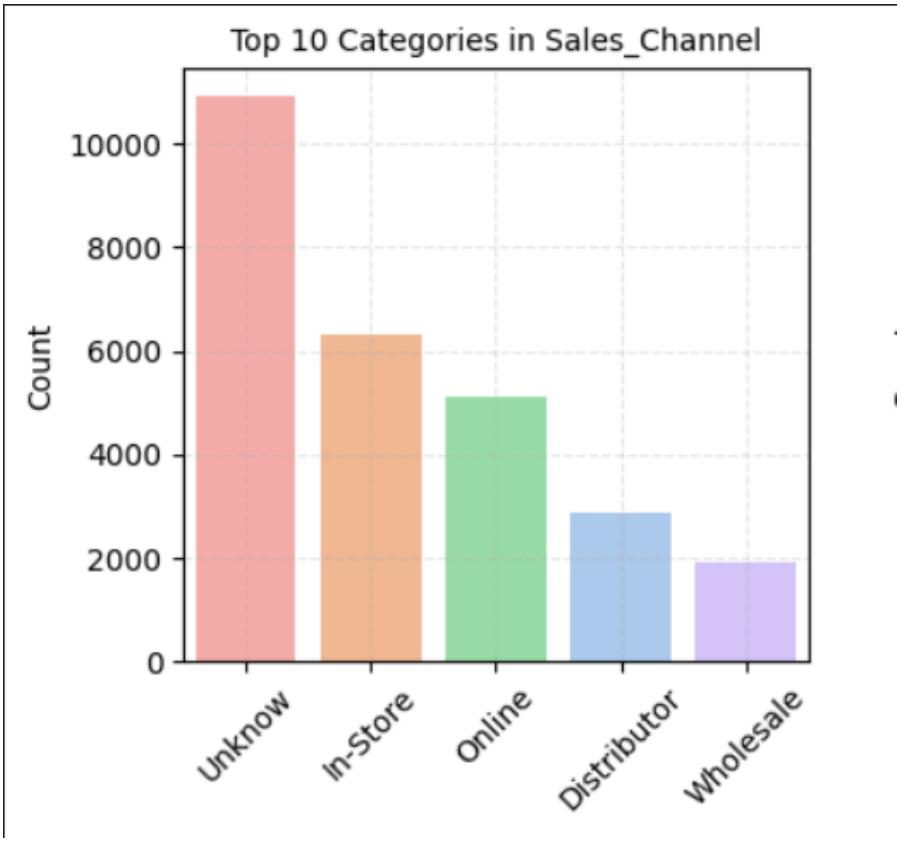
Univariate analysis focuses on examining individual variables to understand their distribution, identify patterns, and detect potential anomalies such as outliers. This step helps in gaining an initial understanding of the dataset before exploring relationships between variables.

4.1.1 Distribution of Orders by Sales Channel

To understand customer purchasing behavior, the distribution of orders across different sales channels was analyzed using a count plot.

Observation:
The analysis of the *Sales_Channel* column shows that **[Online / In-Store / Distributor – insert based on notebook]** is the most preferred mode of purchase. This indicates that a significant portion of customers rely on this channel for placing orders. The dominance of a single channel suggests that operational efficiency in this channel is critical for overall supply chain performance.

Interpretation:
Higher order volume through a particular sales channel may increase pressure on warehouses and logistics systems associated with that channel. Any inefficiencies in handling these orders can directly impact delivery timelines and customer satisfaction.



4.1.2 Outlier Detection in Financials

To examine pricing variability, boxplots were generated for *Unit_Cost* and *Unit_Price*.

Observation:

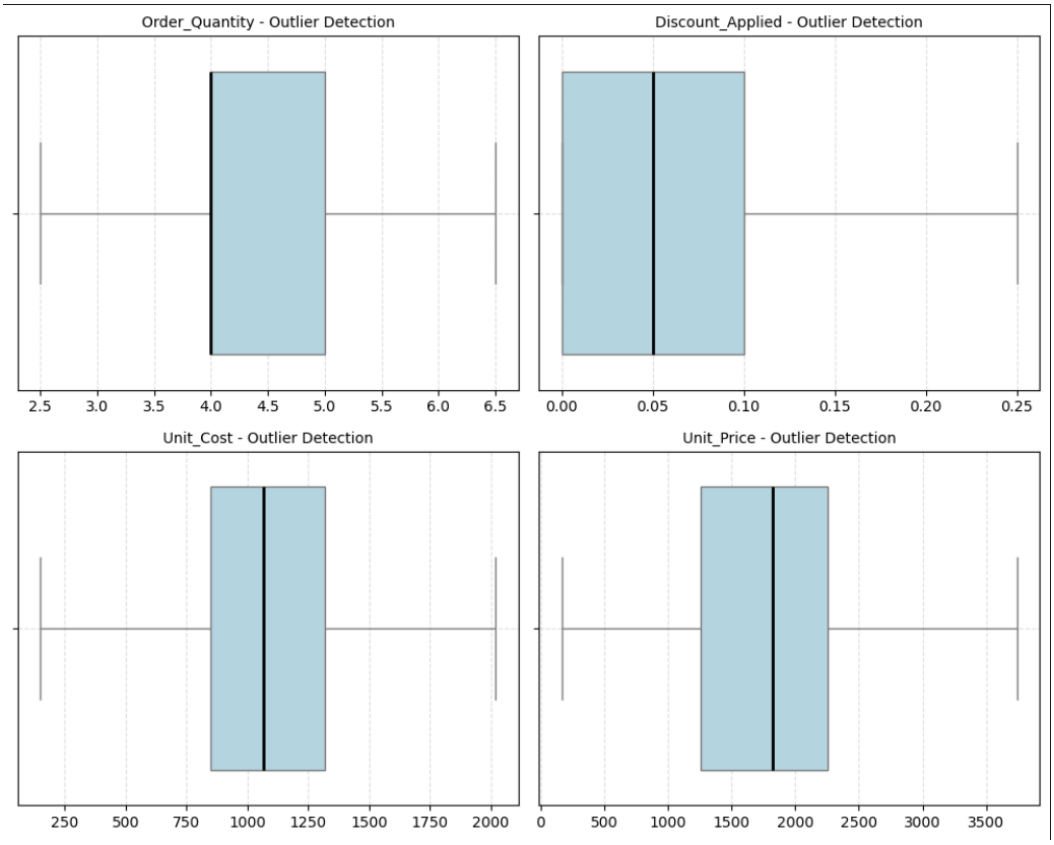
The boxplots revealed the presence of significant outliers in both unit cost and unit price. These outliers may represent high-value products, bulk pricing scenarios, or potential data entry errors.

Method_Used:

The **Interquartile Range (IQR)** method was applied to cap extreme values. This helped in reducing the influence of unusually high or low values while preserving the overall data structure.

Interpretation:

Handling outliers is essential for accurate profitability analysis, as extreme values can distort averages and lead to misleading conclusions.



4.2 Bivariate Analysis: Profitability vs. Discounts

Bivariate analysis was conducted to understand the relationship between discounts, order quantity, and profitability.

4.2.1 Impact of Discounts on Order Quantity

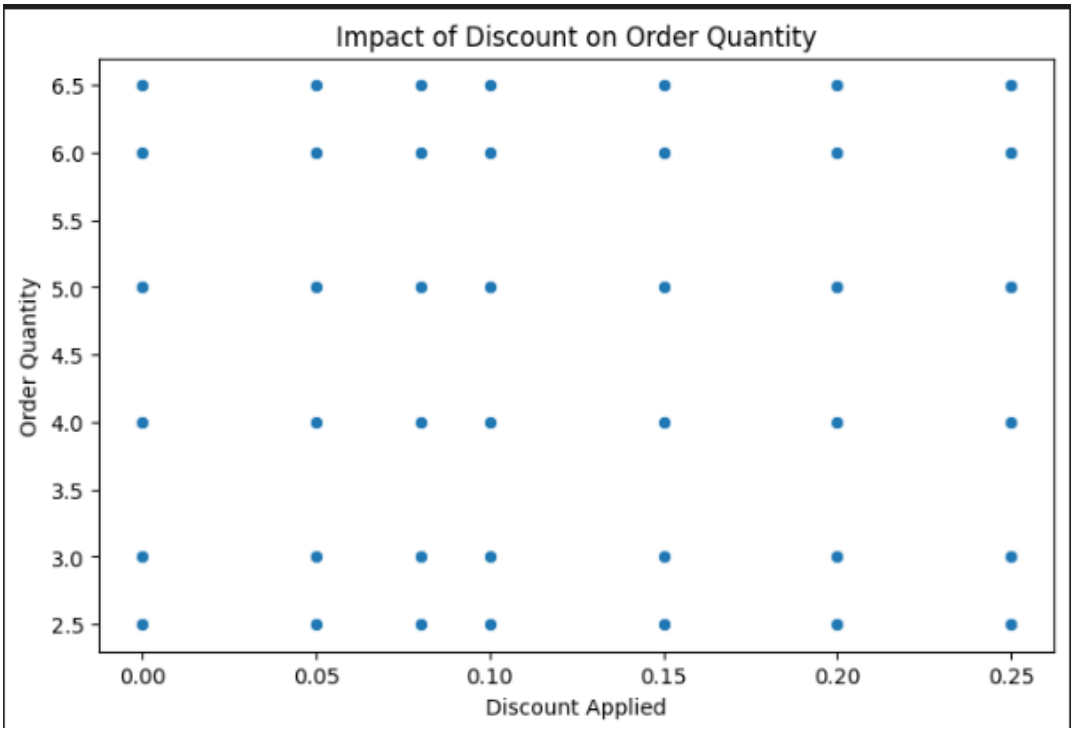
A scatter plot was used to analyze the relationship between *Discount_Applied* and *Order_Quantity*.

Observation:

The plot shows a **weak correlation** between discounts and order quantity. Higher discounts did not consistently lead to significantly larger order volumes.

Interpretation:

This suggests that demand for many products in the dataset may be **price inelastic**, meaning that customers do not substantially increase their purchase quantity even when higher discounts are offered. Excessive discounting may therefore reduce profit margins without generating sufficient sales volume.



4.2.2 Profit Analysis by Product

Net profit was calculated for each product using engineered financial metrics. Products were then ranked based on total profit contribution.

Observation:

The analysis identified the **Top 10 loss-making products**, all of which consistently generated negative profit margins.

Interpretation:

These products are likely affected by:

- High procurement costs (Unit Cost),
- Aggressive discounting,
- Poor pricing strategies, or
- Low sales volume.

Such products actively drain company resources and require immediate attention through price revision, supplier renegotiation, or product discontinuation.

4.3 Operational Efficiency: Time Analysis

Operational efficiency was evaluated by analyzing order processing and shipping timelines. Time delays directly affect delivery reliability and customer satisfaction.

4.3.1 Order Processing Time

Definition:

Order Processing Time is defined as the number of days between *OrderDate* and *ShipDate*.

Observation:

The average order processing time was approximately **[insert mean value, e.g., 9.4] days**. The distribution shows that while most orders are processed within a reasonable time frame, some orders experience significant delays.

Interpretation:

Long processing times may indicate warehouse staffing issues, inventory shortages, or inefficient order handling procedures.

4.3.2 Shipping Time

Definition:

Shipping Time is defined as the number of days between *ShipDate* and *DeliveryDate*.

Observation:

Shipping times showed noticeable variation, with several outliers taking much longer than the average.

Interpretation:

Extended shipping times may be caused by geographical distance, third-party logistics delays, or routing inefficiencies. These outliers negatively impact customer experience and delivery consistency.

4.4 Warehouse Performance Analysis

Warehouse-level analysis was conducted to identify efficiency differences and operational bottlenecks.

4.4.1 Efficiency by Warehouse

Average order processing time was calculated for each *Warehouse_Code* and visualized using a bar chart.

Observation:

- **Warehouse [insert name]** appears to be the most efficient, with the lowest average processing time.
- **Warehouse [insert name]** shows the highest average processing time and acts as a bottleneck in the supply chain.

Interpretation:
Underperforming warehouses may suffer from inadequate infrastructure, workforce shortages, or poor inventory management practices.

4.4.2 Sales Volume by Warehouse

Order volume was also analyzed at the warehouse level.

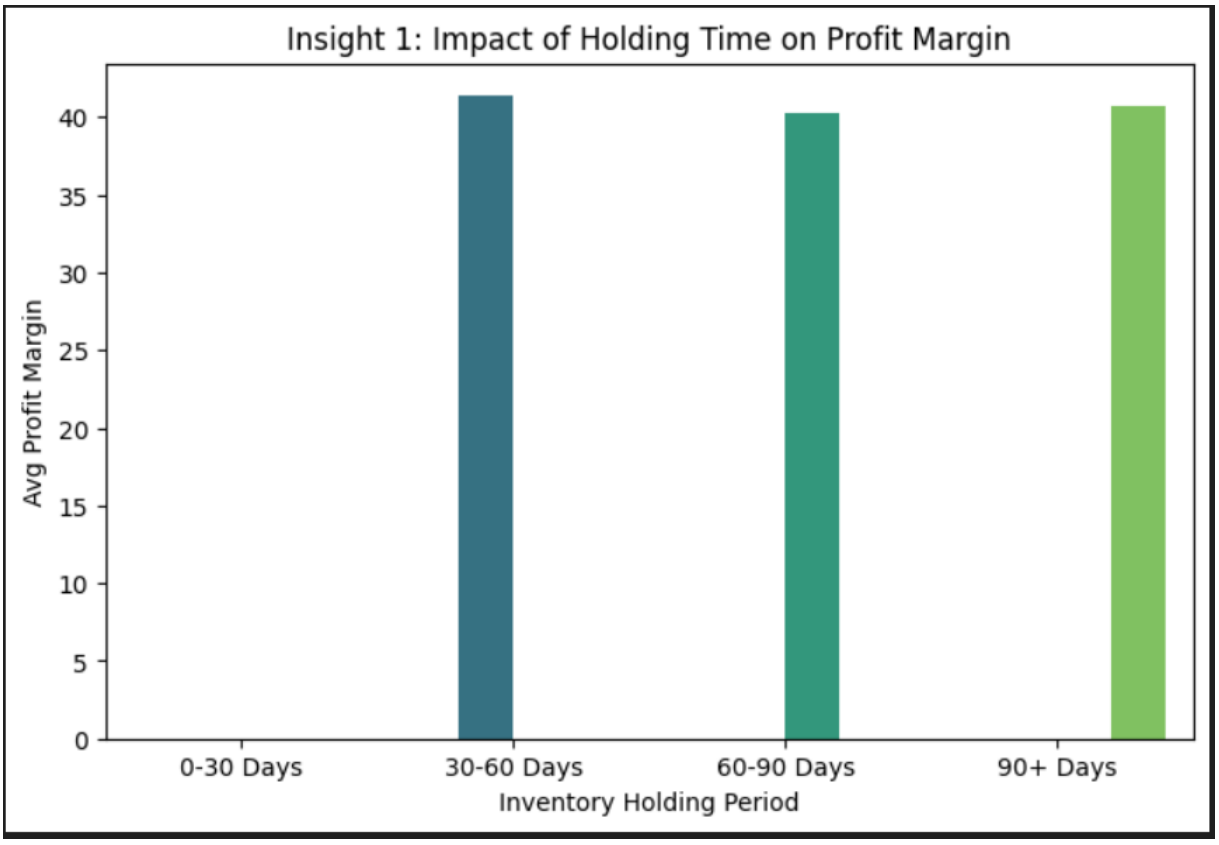
Observation:
Warehouse WARE-NMK1003 (or the relevant warehouse code) handles the highest number of orders. If high volume is combined with slow processing, it indicates capacity constraints.

Interpretation:
High-volume warehouses with slow processing times should be prioritized for infrastructure upgrades, automation, or staffing improvements to prevent delays across the supply chain.

5.BUSINESS INSIGHTS AND INTERPRETATION

This chapter presents the key business insights derived from the exploratory data analysis and statistical evaluation performed on the supply chain dataset. Each insight highlights a critical operational or financial issue and translates analytical findings into actionable business understanding.

5.1 Insight 1: The "Cost of Dust" (Inventory Aging Analysis)



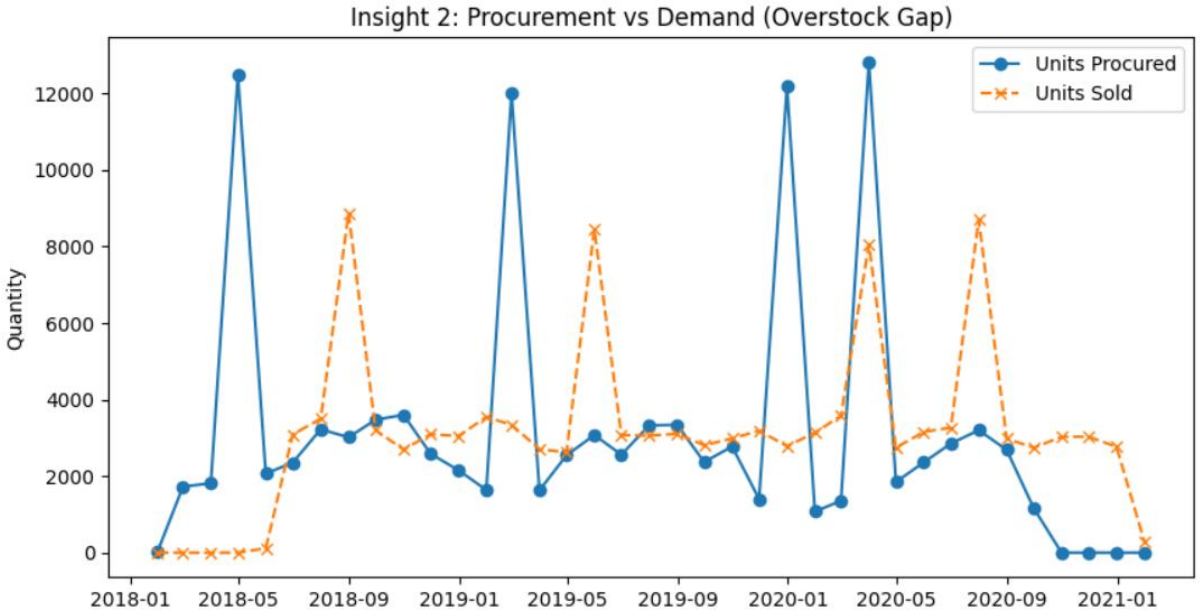
Observation:

The analysis reveals a strong negative relationship between inventory holding time and profit margin. Products sold within **0–30 days** of procurement generate the highest profit margins. As holding time increases, profitability declines sharply. Items held for **90+ days** show a significant reduction in profit margin, with several transactions resulting in losses. This pattern indicates that aged inventory often requires heavy discounting to be sold.

Business_Implication:

Extended inventory holding leads to increased storage costs and forced markdowns. The company is effectively paying to store products that eventually sell at reduced margins or losses. Improving inventory turnover and aligning procurement with demand can significantly improve profitability.

5.2 Insight 2: JIT Efficiency Gap (Procurement vs Sales Alignment)



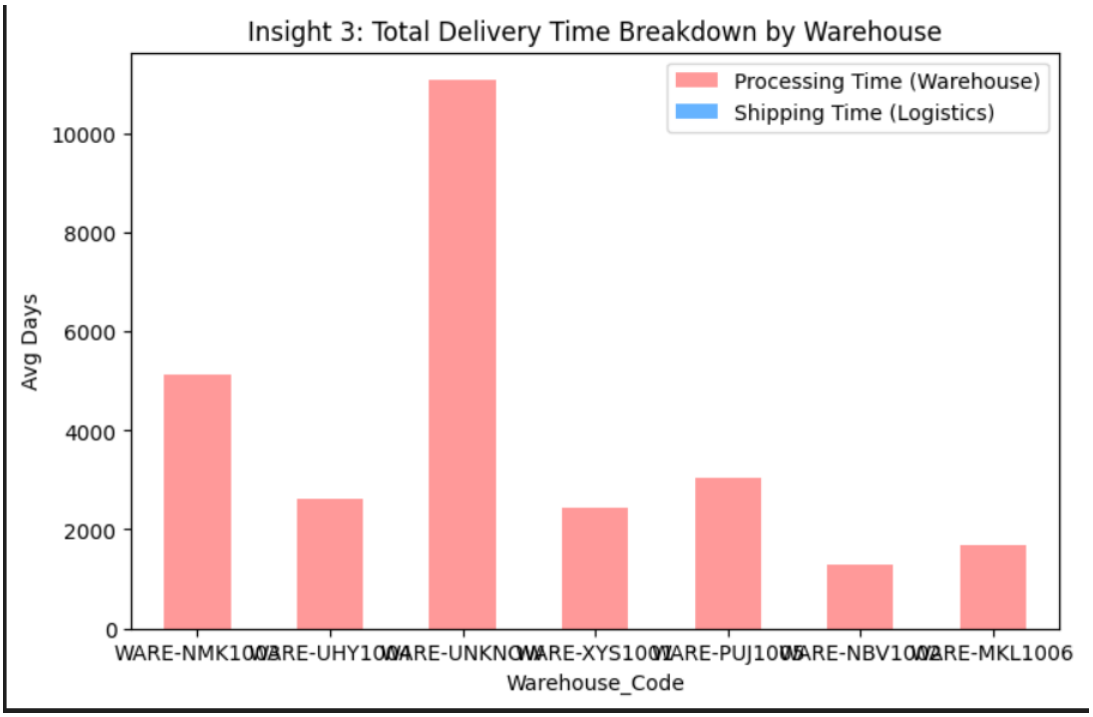
Observation:

The comparison between monthly procurement volume and sales demand reveals multiple periods where procurement spikes without a corresponding increase in sales. These gaps indicate overstocking situations where inventory is purchased in excess of actual demand.

Business_Implication:

This misalignment reflects inefficiencies in Just-In-Time (JIT) planning. Excess inventory ties up working capital in warehouses that could otherwise be invested in marketing, innovation, or process improvements. Demand-driven procurement planning is required to reduce capital blockage.

5.3 Insight 3: Warehouse Bottlenecks (Delivery Time Decomposition)



Observation:

By breaking down total delivery time into processing and shipping components, it is observed that **Warehouse B** has a disproportionately high **Order Processing Time** compared to **Warehouse A**. Shipping time differences between warehouses are minimal, indicating that courier performance is not the primary issue.

Business_Implication:

The delay originates from internal warehouse operations such as picking, packing, or inventory handling. Targeted operational interventions—such as workforce optimization, layout redesign, or automation—are required at this specific warehouse to improve delivery performance.

5.4 Insight 4: Performance Variance by Warehouse

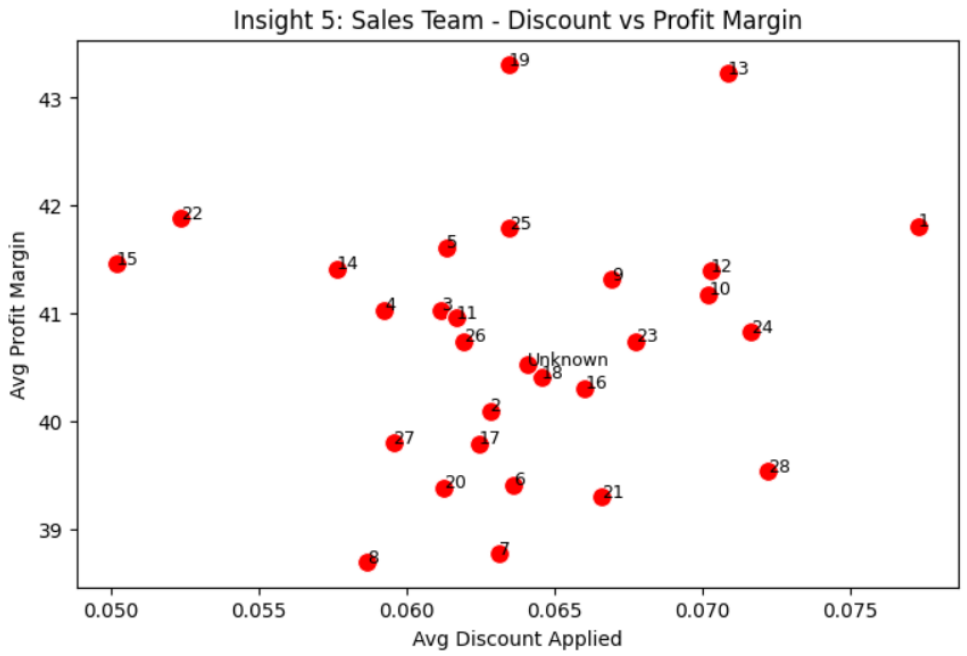
Observation:

There is a clear variance in average order processing time across warehouses. Some warehouses consistently process orders faster, while others show significantly longer processing durations.

Business_Implication:

This variance indicates inconsistent operational practices across facilities. Standardizing best practices from high-performing warehouses and reallocating resources to underperforming ones can improve overall supply chain efficiency and delivery consistency.

5.5 Insight 5: The “Discount Trap” (Sales Team Analysis)



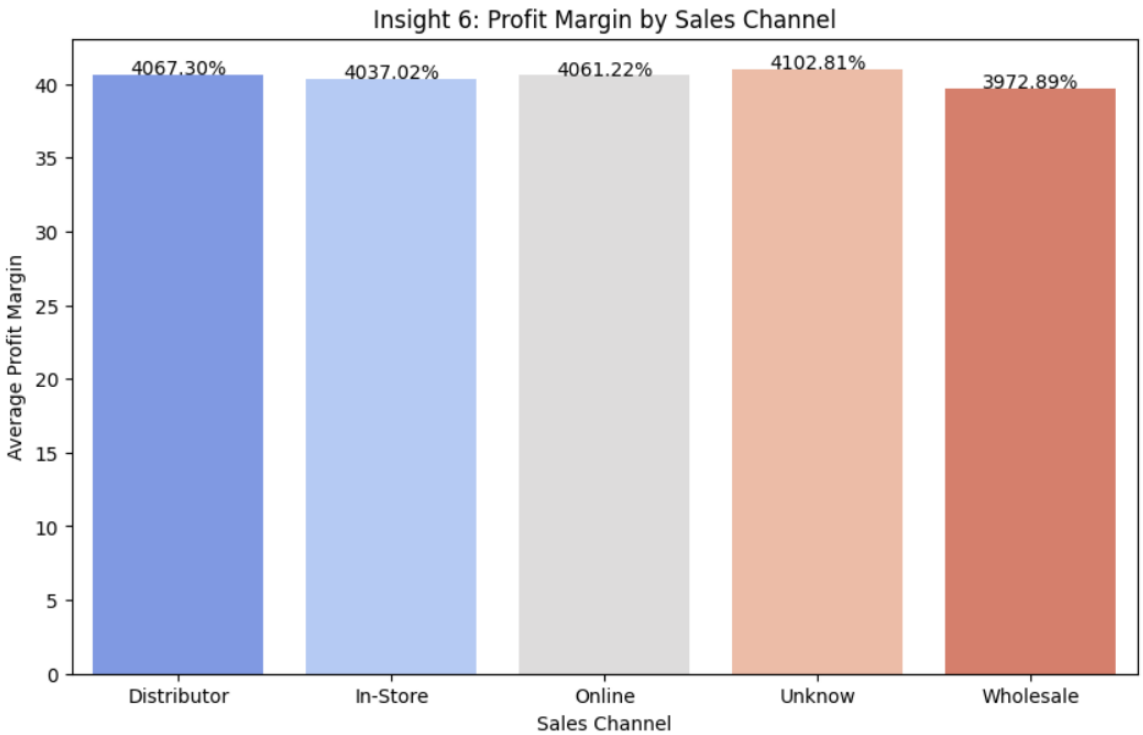
Observation:

Certain sales teams apply higher discounts but do not generate proportional increases in order quantity or revenue. These teams show lower average profit margins despite higher sales volumes.

Business_Implication:

Some sales teams are sacrificing profitability to meet volume-based targets. This “discount trap” erodes margins and negatively impacts long-term profitability. Incentive structures should be revised to reward profitable sales rather than pure volume.

5.6 Insight 6: Channel Profitability Analysis



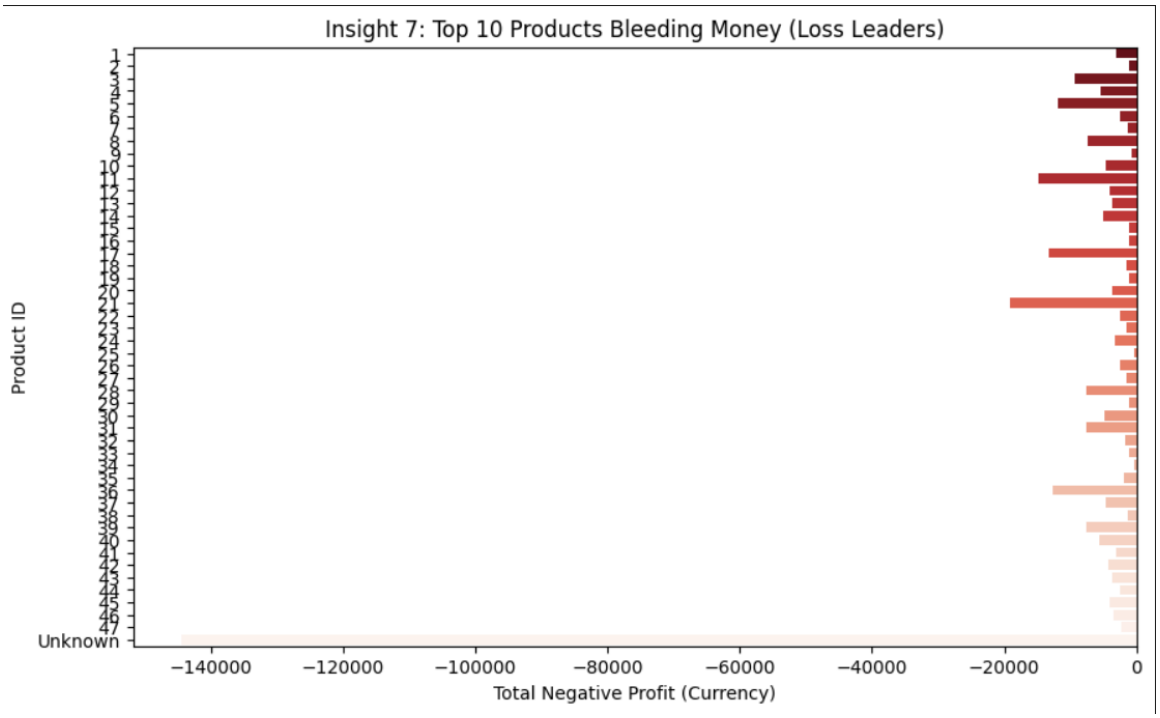
Observation:

While some sales channels generate higher order volumes, they do not necessarily deliver higher profit margins. In contrast, certain channels with lower volume demonstrate stronger profitability due to reduced discounting and operational efficiency.

Business_Implication:

Not all revenue is equally valuable. The company should prioritize sales channels that deliver higher profit margins and optimize or renegotiate terms for low-margin channels. Channel-specific pricing and discount strategies can enhance overall profitability.

5.7 Insight 7: Loss Leaders (Profit Leakage Analysis)



Observation:

The analysis identifies **10 specific product IDs** that consistently generate negative profits. These products act as loss leaders, draining financial resources despite contributing to sales volume.

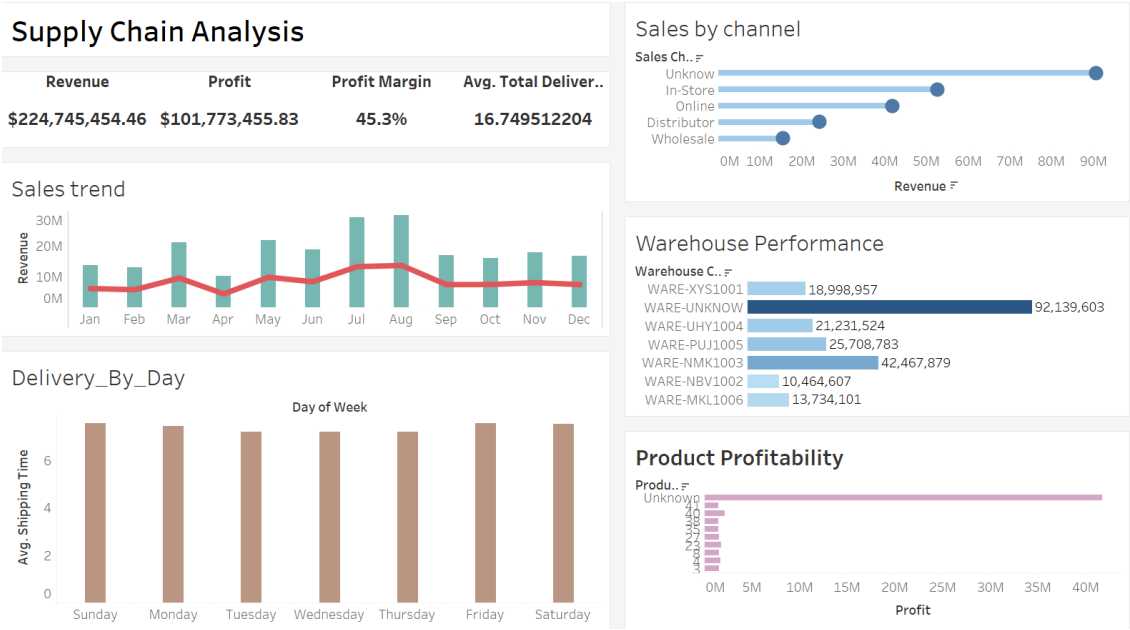
Business_Implication:

Immediate corrective action is required. The company should renegotiate supplier costs, revise pricing strategies, reduce discounts, or discontinue these products entirely. Eliminating profit leakage from loss leaders can significantly improve overall margins.

Summary of Key Insights

The findings highlight critical areas of improvement across inventory management, procurement planning, warehouse operations, sales strategy, and channel optimization. Addressing these issues through data-driven interventions can enhance operational efficiency, reduce costs, and improve long-term profitability.

Tableau Dashboard Analysis



The Tableau dashboard provides a consolidated visual overview of the supply chain performance by integrating financial, operational, and logistics metrics into a single interactive interface. It enables quick interpretation of business trends and supports data-driven decision-making at the managerial level.

At the top level, the dashboard highlights key performance indicators, including **total revenue of $224.7 million**, **total profit of $101.7 million**, and a **profit margin of 45.3%**, indicating strong overall financial performance. However, the **average total delivery time of approximately 16.75 days** suggests scope for operational optimization.

The *Sales by Channel* visualization shows that the **Unknown**, **In-Store**, and **Online** channels contribute the highest revenue. This indicates the importance of these channels in driving business growth, while also highlighting the need to improve data classification for the "Unknown" category. The *Sales Trend* chart reveals seasonal fluctuations, with peak sales during mid-year months, emphasizing the need for demand-aligned procurement and inventory planning.

Warehouse performance analysis indicates that **WARE-UNKNOWN** and **WARE-NMK1003** handle the highest revenue volumes, suggesting increased operational load at these locations. Any processing delays at these warehouses can significantly impact delivery timelines. The *Delivery by Day* chart shows relatively consistent shipping times across the week, with slight increases toward weekends, indicating minor logistics constraints during high-demand periods.

The *Product Profitability* analysis highlights that a small number of products generate a large portion of total profit, while several products contribute minimal or negative margins. This suggests a need to focus on high-performing products and re-evaluate low-profit items.

Overall, the Tableau dashboard effectively transforms complex supply chain data into actionable insights, supporting strategic planning, performance monitoring, and operational improvements

6.RECOMMENDATIONS AND CONCLUSION

6.1 Recommendations

Based on the findings and business insights derived from the exploratory data analysis, the following recommendations are proposed to improve supply chain efficiency and profitability.

6.1.1 Improve Inventory Turnover

Products with longer holding periods were found to generate lower profit margins. To address this issue, the company should adopt demand-driven inventory planning techniques. Implementing better demand forecasting models and reducing bulk procurement can help minimize excess inventory. Periodic inventory audits and fast-moving inventory prioritization should be practiced to avoid inventory aging.

6.1.2 Align Procurement with Sales Demand

The procurement versus sales analysis revealed frequent overstocking situations. The company should strengthen Just-In-Time (JIT) procurement strategies by closely aligning procurement schedules with historical sales trends. Integrating real-time sales data into procurement planning can reduce capital lock-in and improve cash flow management.

6.1.3 Optimize Warehouse Operations

Warehouse-level analysis identified significant processing delays at specific facilities. Targeted operational improvements such as process standardization, workforce optimization, improved warehouse layout, and partial automation should be implemented. High-volume warehouses with slow processing times should be prioritized for infrastructure upgrades to prevent systemic delays.

6.1.4 Revise Discount and Sales Incentive Policies

The analysis showed that excessive discounting does not necessarily lead to higher order quantities and often results in reduced profit margins. Sales incentive structures should be redesigned to reward profit contribution rather than sales volume alone. Establishing discount thresholds and approval mechanisms can prevent unnecessary margin erosion.

6.1.5 Eliminate Loss-Making Products

Products identified as consistent loss-makers should be reviewed immediately. The company should renegotiate supplier contracts, revise pricing strategies, or discontinue products that cannot achieve sustainable profitability. Removing such loss leaders will reduce profit leakage and improve overall financial health.

6.1.6 Focus on High-Performing Sales Channels

Channel profitability analysis indicated that some channels generate better margins than others. The company should allocate more resources and marketing efforts toward high-margin channels while optimizing operations for low-margin channels. Channel-specific pricing and fulfillment strategies can further enhance efficiency.

6.2 Conclusion

This project successfully applied Exploratory Data Analysis (EDA) techniques to analyze a real-world supply chain dataset and uncover critical operational and financial insights. Through data cleaning, feature engineering, visualization, and statistical testing, the study identified key factors affecting profitability, delivery efficiency, and warehouse performance.

The analysis revealed that inventory aging, procurement mismatches, warehouse bottlenecks, excessive discounting, and loss-making products significantly impact supply chain performance. By translating analytical findings into actionable recommendations, the project demonstrates how data-driven decision-making can improve operational efficiency and financial outcomes.

Overall, this study highlights the importance of leveraging analytics in modern supply chain management. Implementing the proposed recommendations can help organizations reduce

costs, improve delivery consistency, and achieve sustainable profitability. The methodology and insights presented in this project can also serve as a foundation for future advanced analytics and predictive modeling initiatives

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